NEW PROJECT · STORAGE SIZING & RAID DESIGN

CCTV Storage Design & Capacity Planning

500 × 4 MP cameras across five Hybrid SAN storage servers — bitrate & capacity calculation, per-server HDD configuration and retention planning.

500 4 MP Cameras

5 Storage Servers

100 cams / server

H.265 @ 4 Mbps

24/7 Continuous

RAID-5 / 6

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DESIGN BASIS

Bitrate & Capacity Calculation

H.265 · 4 Mbps · 24/7

DESIGN ASSUMPTIONS

Camera resolution	4 MP (2560 × 1440)
Codec	H.265 (HEVC)
Main-stream bitrate	**4 Mbps** CBR
Frame rate	25 fps
Recording mode	24 / 7 continuous
Cameras per server	**100**
Storage servers	**5** (= 500 cameras)
Retention (baseline)	**30 days** — scalable

Storage / day (GB) = Bitrate (Mbps) × **`10.8`** → 4 × 10.8 = **`43.2 GB/day`**

CAPACITY REQUIREMENT

PER CAMERA

0.5 MB/s | **43.2** GB/day | **~1.3** TB / 30 d

PER STORAGE SERVER — 100 CAMERAS

400 Mbps write | **4.32** TB/day | **~130** TB / 30 d

WHOLE SYSTEM — 5 SERVERS / 500 CAMERAS

21.6 TB/day | **~648** TB / 30 d usable

Add ~15% operating headroom for index, overhead & burst → design target ≈ **150 TB usable per server**. At 6 Mbps multiply ×1.5; using H.264 instead of H.265 roughly doubles all figures.

HARDWARE CONFIGURATION

Per-Server HDD Build & Retention

× 5 identical units

RECOMMENDED PER-SERVER CONFIGURATION

Hikvision Hybrid SAN — 24-Bay

`DS-A81024S  (records + stores its own 100 cameras)`

Hard drives	24 × 8 TB surveillance HDD
Drive type	Seagate SkyHawk AI / WD Purple Pro
RAID layout	2 × RAID-5 (10 data + 1 parity each)
Hot spares	2 × global
Raw capacity	192 TB
Usable capacity	**≈ 160 TB** (~152 TB after FS)
Retention delivered	**≈ 35 days** @ 100 cams / 4 Mbps

System total: 5 × 192 TB = **960 TB raw** · ≈ 760 TB usable · ≈ 108 TB/day aggregate write across the five units.

DRIVE SIZE → RETENTION (PER SERVER, 100 CAMS @ 4 MBPS)

DRIVE SIZE	USABLE / SERVER	RETENTION
8 TB baseline	≈ 152 TB	**≈ 35 days**
10 TB	≈ 190 TB	≈ 44 days
12 TB	≈ 228 TB	≈ 53 days
16 TB	≈ 304 TB	≈ 70 days
20 TB	≈ 380 TB	≈ 88 days
24 TB	≈ 456 TB	≈ 105 days

Usable based on 2 × RAID-5 + 2 hot-spares (20 data disks) × 0.95 filesystem factor

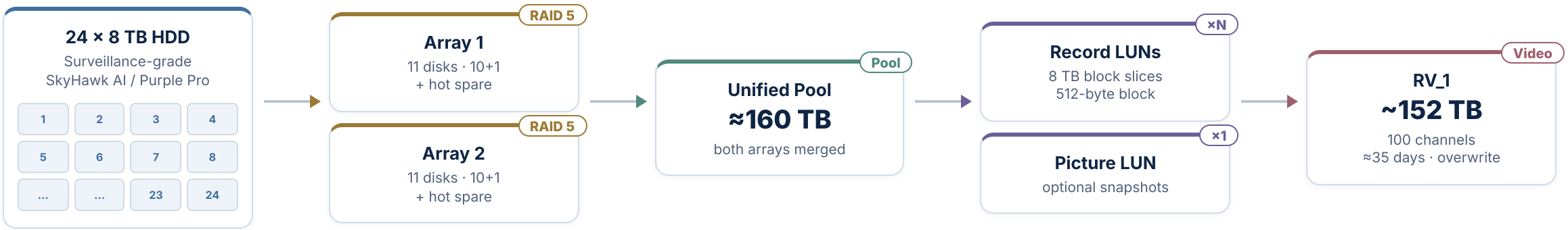
RAID level: RAID-5 is fine up to 10 TB drives. For **12 TB and above use RAID-6** (dual parity) — large-drive rebuilds are long and RAID-6 survives a second failure mid-rebuild.

Throughput check: 100 × 4 Mbps = 400 Mbps (~50 MB/s) write per unit — comfortably inside a 24-disk array and dual 10 GbE uplinks.

PHYSICAL → LOGICAL

Storage Stack — per Hybrid SAN Unit

Disks → RAID → Pool → LUN → Record Volume



Physical disks — raw drives · RAID — redundancy (survives 1 disk / array) · Pool — arrays merged · LUN — logical slices to HikCentral · Record Volume — where video writes

Replicated across all 5 units. Each Hybrid SAN is an identical, self-contained recording + storage node managed by HikCentral Professional — 100 cameras each — so the system scales linearly and a single unit's failure affects only its own 100 channels.